

Ex. No. 9 – Use Process Explorer to Identify Suspicious Processes

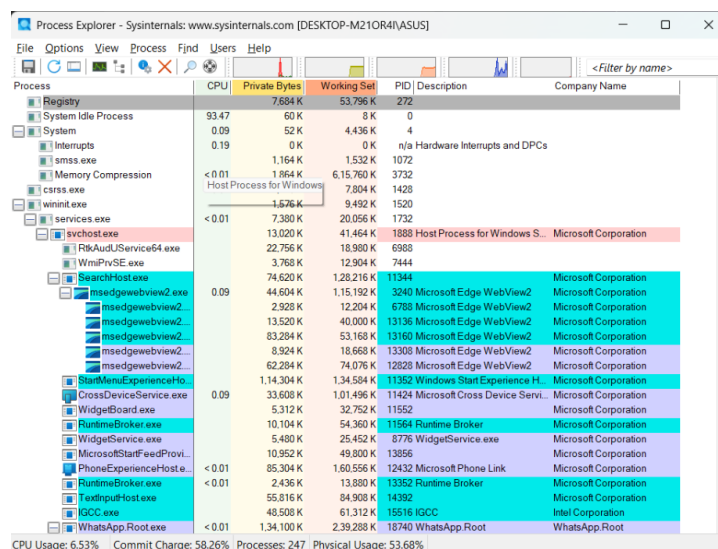
Aim: To use Microsoft Sysinternals Process Explorer to examine running Windows processes and identify potentially suspicious processes by checking process details, executable path, digital signature, registered services, CPU usage, and memory usage.

Tools Required: Windows 10/11; Microsoft Sysinternals Process Explorer.

Procedure

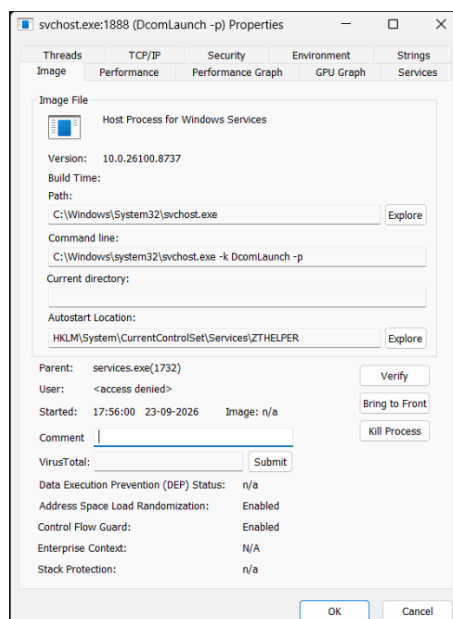
Step 1: Launch Process Explorer

Microsoft Sysinternals Process Explorer was downloaded and extracted. The procexp64.exe application was opened to display the processes currently running on the Windows system.



Step 2: View Running Processes

The main Process Explorer window displayed active processes in a hierarchical process tree. The available information included Process Name, CPU Usage, Private Bytes, Working Set, Process ID (PID), Description, and Company Name.



Step 3: Select a Process

The svchost.exe process with PID 1888 was selected for detailed examination. The process was displayed under the parent process services.exe.

Step 4: Check Process Properties

The Properties window of svchost.exe was opened to examine its process information.

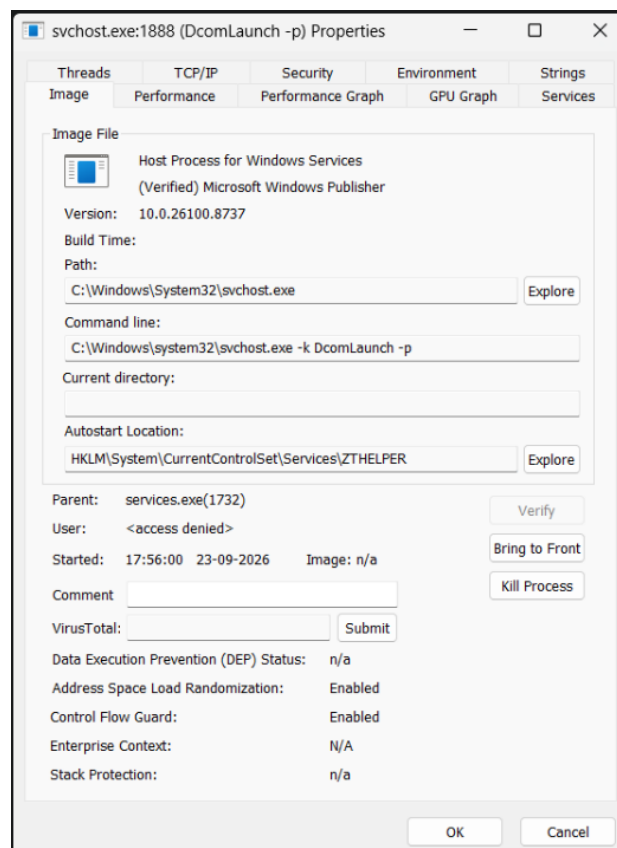
Process	svchost.exe
PID	1888
Description	Host Process for Windows Services
Version	10.0.26100.8737
Path	C:\Windows\System32\svchost.exe
Parent	services.exe (1732)

Command Line: C:\Windows\system32\svchost.exe -k DcomLaunch -p

Ex. No. 9 – Process Examination (Continued)

Step 5: Verify Digital Signature

The digital signature of the selected process was checked using the Verify option in the Properties window. The result displayed “(Verified) Microsoft Windows Publisher”, indicating that the executable was verified as being published by Microsoft Windows.

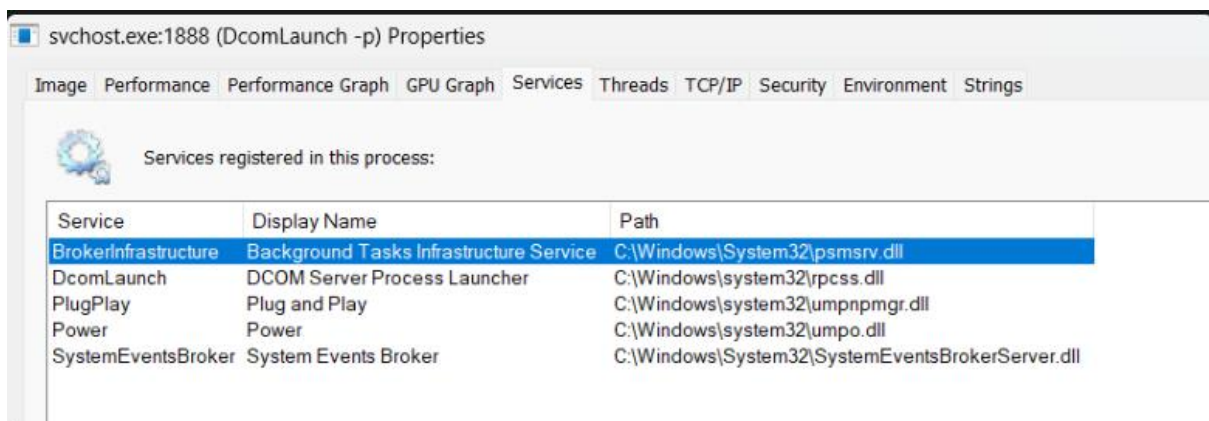


Step 6: Check Registered Services

The Services tab was opened to examine the services associated with the selected svchost.exe process.

Service	Display Name
BrokerInfrastructure	Background Tasks Infrastructure Service
DcomLaunch	DCOM Server Process Launcher
PlugPlay	Plug and Play
Power	Power
SystemEventsBroker	System Events Broker

The associated service paths were located under the Windows System32 directory.



Step 7: Observe CPU and Memory Usage

The Process Explorer main window was examined to observe the resource usage of the selected process. The approximate values recorded for svchost.exe were:

CPU	<0.01%
Private Bytes	13,024 K
Working Set	41,516 K
PID	1888
Company Name	Microsoft Corporation

Process Explorer - Sysinternals: www.sysinternals.com [DESKTOP-M210R4\ASUS]

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Registry	9.704 K	55.684 K	272			
System Idle Process	97.31	60 K	8 K	0		
System	0.66	60 K	7.380 K	4		
Interrupts	0.28	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1.164 K	1.540 K	1072		
Memory Compression		1.940 K	7.92.732 K	3732		
csrss.exe	< 0.01	2.536 K	7.864 K	1428		
wininit.exe		1.576 K	9.948 K	1520		
services.exe		7.580 K	20.640 K	1732		
svchost.exe	< 0.01	13.952 K	44.428 K	1888	Host Process for Windows S...	Microsoft Corporation
RtkAudUService64.exe		22.780 K	19.624 K	6988		
WmiPrivSE.exe		3.592 K	13.316 K	7444		
SearchHost.exe		74.776 K	1.58.032 K	11344		Microsoft Corporation
msedgewebview2.exe		44.580 K	1.16.856 K	3240	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		2.940 K	12.764 K	6788	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		13.556 K	41.092 K	13136	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		83.260 K	71.316 K	13160	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		8.940 K	19.464 K	13308	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		68.000 K	93.292 K	12828	Microsoft Edge WebView2	Microsoft Corporation
StartMenuExperienceHo...		1.14.532 K	1.82.076 K	11352	Windows Start Experience H...	Microsoft Corporation
CrossDeviceService.exe	< 0.01	33.832 K	1.02.716 K	11424	Microsoft Cross Device Servi...	Microsoft Corporation
WidgetBoard.exe		5.192 K	32.988 K	11552		Microsoft Corporation
RuntimeBroker.exe		10.208 K	56.224 K	11564	Runtime Broker	Microsoft Corporation
WidgetService.exe		6.132 K	30.124 K	8776	WidgetService.exe	Microsoft Corporation
MicrosoftStartFeedProvi...		12.372 K	51.972 K	13856		Microsoft Corporation
PhoneExperienceHoste...		86.172 K	1.64.832 K	12432	Microsoft Phone Link	Microsoft Corporation
RuntimeBroker.exe		2.432 K	14.144 K	13352	Runtime Broker	Microsoft Corporation
TextInputHost.exe		55.432 K	1.11.612 K	14392		Microsoft Corporation
IGCC.exe		48.536 K	61.764 K	15516	IGCC	Intel Corporation
WhatsApp.Root.exe	< 0.01	1.38.368 K	2.62.704 K	18740	WhatsApp.Root	WhatsApp Root

CPU Usage: 1.97% Commit Charge: 64.20% Processes: 244 Physical Usage: 58.31%

Ex. No. 9 – Process Analysis and Result

Step 8: Analyze the Process

The selected svchost.exe process was analyzed using its process name, executable path, parent process, digital signature, company name, registered services, CPU usage, and memory usage. The executable was located at C:\Windows\System32\svchost.exe and was verified as Microsoft Windows Publisher. The process was also associated with standard Windows services. Based on the characteristics observed during the examination, no obvious suspicious indicators were identified.

Step 9: Process Control Options

The context menu of the selected process was opened. The available options included Kill Process, Kill Process Tree, Restart, Suspend, and Create Dump.

System	0.19	52 K	4.452 K	4	
Interrupts	0.56	0 K	0 K	n/a	Hardware Interrupts and DPCs
smss.exe		1.164 K	1.532 K	1072	
Memory Compression		1.860 K	6.60.900 K	3732	
csrss.exe	< 0.01	2.520 K	7.744 K	1428	
wininit.exe		1.576 K	9.828 K	1520	
services.exe		7.224 K	19.992 K	1732	
svchost.exe		12.860 K	41.408 K	1888	Host Process for Windows S...
RtkAudUService64.exe		22.756 K	18.980 K	6988	
WmiPrivSE.exe		3.672 K	13.348 K	7444	
SearchHost.exe		74.628 K	1.28.476 K	11344	
msedgewebview2.exe		44.576 K	1.15.692 K	3240	Microsoft Edge WebView2
msedgewebview2...		2.928 K	12.204 K	6788	Microsoft Edge WebView2
msedgewebview2...		13.448 K	40.368 K	13136	Microsoft Edge WebView2
msedgewebview2...		83.280 K	53.164 K	13160	Microsoft Edge WebView2
msedgewebview2...		8.924 K	18.668 K	13308	Microsoft Edge WebView2
msedgewebview2...		62.284 K	74.056 K	12828	Microsoft Edge WebView2
StartMenuExperienceHo...		1.14.264 K	1.34.836 K	11352	Windows Start Experience H...
CrossDeviceService.exe	0.09	33.596 K	1.01.528 K	11424	Microsoft Cross Device Ser...
WidgetBoard.exe		5.184 K	32.664 K	11552	

Window

Set Affinity...

Set Priority

Kill Process Del

Kill Process Tree Shift+Del

Restart

Suspend

Create Dump

The Kill Process option was not used because the examined svchost.exe process did not show any obvious suspicious indicators. It had a verified Microsoft Windows publisher signature, was located in the Windows System32 directory, and was associated with standard Windows services. Therefore, terminating the process was not necessary for this experiment.

Observation

The svchost.exe process with PID 1888 was examined using Microsoft Sysinternals Process Explorer. Its process details, executable path, digital signature, parent process, registered services, CPU usage, and memory usage were checked. The process was located in the Windows System32 directory and was verified as Microsoft Windows Publisher.

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

Microsoft Sysinternals Process Explorer was successfully used to examine a running Windows process and identify potential suspicious indicators. The examined svchost.exe process showed characteristics consistent with a legitimate Windows system process, and no obvious suspicious indicators were observed during the examination.